

Response to Referee 1

Manuscript JFM-2026-1781

Strain-coupled one-dimensional turbulence for rapid distortion

Sharma & Kerstein

We thank the referee for a careful reading and for identifying the two directions in which the original manuscript was underdeveloped. The paper has been substantially rewritten. It is now built around the single-line analysis the referee identified as the strongest part; an independent strained-box DNS with exact-RDT companions has been added as the spectral benchmark of record; the leading-edge application is carried through to an absolute prediction and compared with four experiments; and the closure deficiencies the referee named are now measured against the DNS and stated as quantified limits rather than left open. Section, figure, table and equation numbers below refer to the revised manuscript.

Major point 1 (validation of the main result outside the model).

“The central claim is that at $Sk_t/\varepsilon \approx 0.8$ the spectrum develops a broadband distortion that differs from the rigid spectral translation predicted by rapid-distortion theory. This is established by comparing the model with itself, once with the strain switched on and once with it switched off. ... the predicted distorted spectrum is never directly compared against an independent spectral benchmark.”

Response. We accept the objection. The strain-on/strain-off comparison is retained only as an internal decomposition (now § 4.3, figure 9) and is no longer offered as validation. The validation of record is now an independent, projection-matched spectral benchmark: a 128^3 strained-box DNS of plane-strained homogeneous turbulence in a deforming-frame pseudo-spectral formulation, together with exact-RDT companion calculations, all reduced to the same one-dimensional line observables (§ 4.6; DNS protocol in Appendix B.3). The three-way comparison of the strain-induced spectral anisotropy—SC-ODT against exact projected RDT against DNS—is figure 11 and table 1 at $e = 1$, $Sk_t/\varepsilon = 0.8$. It shows directly what the model reproduces (the upwash amplification) and where it departs (its wavenumber-uniform signature against the wavenumber-dependent DNS and RDT). Section 4.6.2 and figure 12 add a distance-from-linear-theory envelope (the rms log-residual of a two-parameter exact-RDT-distorted von Kármán fit) as a function of strain rapidity Sk_t/ε , giving the departure a measured, monotone size. A five-way version at three rapidities ($Sk_t/\varepsilon = 0.4, 0.8, 16$) is figure 16.

Major point 2 (the original single-line development is not used in the reported calculations).

“The simulations are performed using the closure introduced in Section 3.2, whereas the more complete single-line formulation developed later is not used in generating the principal results. ... it is unclear whether the main spectral conclusions would remain unchanged under the formulation that the paper ultimately argues is the more appropriate one.”

Response. The referee is correct that the reported spectral calculations use the algebraic (LRR) moment closure, and we now state this plainly at each figure and in Section 5.7. Section 5 has been reframed so that it no longer claims a closed single-line functional that then goes unused. Instead it (i) states the obstruction exactly—the line carries the transverse-integrated spectra of equation (5.3), and the rapid term requires the direction-resolved projection weight that those integrals destroy (§ 5.2); (ii) derives sharp bounds on the rapid term from line data alone by linear programming, with the residual indeterminacy tabulated (§ 5.4, table 2, Appendix A.4); and (iii) shows that the collapsed spectral form of equations (5.6)–(5.9) reduces exactly to the moment closure in the isotropic limit, equation (5.11), which is what justifies using that closure in the simulations (§ 5.7, figure 14). The consistency question the referee raises—whether the conclusions survive where the moment closure loses accuracy—is answered by measurement rather than assumption: the axisymmetry hypothesis on which the collapse rests is measured against the DNS (§ 5.5, table 3, figure 13) and the component-ordering error is localised (§ 5.6; next point), while the envelope of § 4.6.2 bounds the resulting distance from linear theory over the strain range. Exercising the full wavenumber-dependent kernel of equation (5.6) inside ODT is identified as the missing ingredient (§ 6.4) and left as future work; we do not claim it here.

Major point 3 (the rapid pressure–strain closure becomes qualitatively inaccurate under plane strain).

“Under plane strain, the exact solution predicts growth of the neutral spanwise component, which eventually overtakes the upwash component, whereas both closures predict the opposite trend. ... What remains unclear is how sensitive the subsequent spectral predictions are to this known deficiency.”

Response. This is now measured, localised and bounded rather than only acknowledged. The exact rapid term evaluated in the DNS (§ 5.6) gives $\Pi_{33}^{(r)}/k_t S = 0.14$ at $Sk_t/\varepsilon = 0.8$ and 0.27 at 16, the latter exceeding $\Pi_{11}^{(r)} = 0.23$: the overtaking is real and sets in at rapid strain. We show that the error lives in the moment closure’s anisotropy-dependent term—LRR reproduces $\Pi_{33}^{(r)}$ at $Sk_t/\varepsilon = 0.8$ but delivers half the exact value at 16, and IP gives zero identically—and not in the line model: the ODT moment-level b_{ij} follows the closure to $\sim 2\%$. In the spectral representation the ordering is carried by the anisotropy scalar c through the kernels (5.7)–(5.9), which the moment reduction discards. The same deficiency appears at the moment level in the plane-strain b_{11} (§ 3.4, figure 6): the fast-strain DNS tracks exact RDT while the closures give $b_{11} \approx 0$. On sensitivity: the acoustically relevant quantity is the upwash amplification b_{22} , and the A1 control experiment (figure 13b, table 3) shows it indistinguishable between plane and axisymmetric strain even at $Sk_t/\varepsilon = 16$, i.e. robust to the azimuthal structure that carries the ordering error; the distance envelope of § 4.6.2 gives the overall sensitivity of the spectral prediction its measured bound.

Major point 4 (the leading-edge connection is not carried through).

“First, it is not demonstrated that the spectrum produced by the ODT line is the same quantity required by Amiet’s theory, which is naturally formulated in terms of chordwise and spanwise wavenumbers. Second, the strain history responsible for generating the turbulence distortion is deferred to future work.”

Response. The connection is now carried through in a dedicated Section 7. The two-dimensional upwash spectrum $\Phi_{ww}(K_x, K_y)$ that Amiet’s theory consumes is supplied from the axisymmetric representation of § 5.3 fitted to the line spectra, so the model’s one-dimensional output is

mapped to the quantity the aeroacoustic kernel requires; the computed distorted inflow is then fed to Amiet’s response both as a ratio from which aerofoil geometry and observer position cancel (figure 17) and as an absolute far-field prediction at the reference operating point (figure 18). The assembled chain is compared with four published experiments (figure 19) and lies within 1–2 dB rms where the facilities are cleanest; the term omitted by the frozen-spectrum assumption is 5–20 dB (figures 17–21). On the second part, we agree the strain history from a real stagnation flow is not modelled, and we now scope the claim as a frozen-strain calculation: the operating point is entered through an effective strain estimated geometrically (potential-flow stagnation-line accumulation truncated at the eddy scale, varied over $\Lambda/4$ – Λ ; figure 19 caption), and the distance envelope of § 4.6.2 is carried through as the uncertainty statement (figures 20–21). The mapping from model operating point to the stagnation environment is stated as the remaining step, not claimed as solved.

Minor points.

(i) Novelty of pre-impact distortion; dos Santos, Ribeiro, Piccolo. Response. The introduction has been rewritten. The pre-impact distortion is now presented as established, with references to dos Santos *et al.* (2023), Ribeiro *et al.* (2023) and Piccolo *et al.* (2024), and the distinction our contribution rests on—prescribing an RDT- or measurement-derived spectrum versus *evolving* one, including the nonlinear relaxation—is stated explicitly (§ 1).

(ii) The dissipation estimate as the residual of two larger terms. Response. The residual-of-two-large-terms estimate has been replaced. Because the eddy events conserve energy and the production is exact, molecular dissipation is the only sink and is fixed by the resolved energy budget of equation (4.1), with every term available on the line (§ 4.4). This estimator-free budget gives $\chi = Sk_t/\varepsilon \approx 1.2$, corroborated within a few tens of per cent by the direct gradient estimate $\langle 2\nu(\partial u_i/\partial y)^2 \rangle \approx 1.1$; the spectral peak migration corroborates the $\chi = O(1)$ reading independently of the dissipation estimate altogether.

(iii) The Lee & Reynolds strain-parameter identification (factor of two). Response. This is now discussed where the comparison is made (§ 3.4). Lee & Reynolds characterise each run by $S^* = Sq^2/\varepsilon$, which equals 2χ *identically* because $q^2 = 2k_t$; the manuscript now states that this is a correspondence of definitions, not a calibration, and notes that the axisymmetric anisotropy they report is essentially independent of S^* , so the comparison target is parameter-free.

(iv) Propositions 1 and 2. Response. Agreed. They are demoted from results to consistency checks and presented as such in the verification section (§ 3.1–3.2).

(v) Length; Section 2 non-original; implementation detail to supplementary material. Response. The standard-ODT material of § 2.1 is compressed to a review keyed to citations of the original ODT papers, and established-versus-new material is separated through Section 2. Implementation detail, the CMK comparison (§ 4.5 summary; full account in supplementary § S1) and the full intervention catalogue (supplementary § S2; labels in Appendix B.1) have been moved out of the main text.

We are grateful for the recommendation to build the study around the single-line analysis; that is now the organising spine of the revised paper.